Draft

Weekly Problem Sets

Each week includes **10 theoretical problems** and **10 coding problems** based on **real-world datasets**.

Datasets are chosen from:

- UCI Machine Learning Repository
- OpenML
- R datasets (ISLR, MASS, etc.)

Week 2: Regularization Methods

Theory Problems

1. Explain why multicollinearity leads to unstable OLS estimates.
2. Write the ridge regression optimization problem

$$\min_{\beta}\{RSS + \lambda \sum_{j=1}^{p} \beta_j^2\}$$

Explain how λ affects coefficient shrinkage.

3. Explain why ridge regression reduces variance but increases bias.
4. Explain why the lasso penalty

$$\sum_{j=1}^{p} |\beta_j|$$

can produce sparse solutions.

5. Compare ridge regression and lasso in terms of interpretability.
6. Explain why predictors should be standardized before applying ridge or lasso.
7. Describe how cross-validation selects λ.
8. Explain the geometric difference between L_1 and L_2 penalties.
9. Suppose $\lambda = 0$. What model is obtained?
10. Explain why lasso is useful when $p > n$.

Coding Problems

Datasets: - Hitters (ISLR) - Wine Quality (UCI)

1. Fit an OLS model predicting salary in the Hitters dataset.
2. Fit ridge regression and plot coefficient paths.
3. Fit lasso regression and identify selected variables.
4. Use cross-validation to select optimal λ.
5. Compare ridge vs lasso prediction error.
6. Standardize predictors and refit ridge regression.
7. Compare coefficients between OLS and ridge.
8. Repeat analysis using the Wine dataset.
9. Plot prediction error vs λ.
10. Interpret which variables are most important.

Week 3: Flexible and Nonparametric Models

Theory Problems

1. Describe the KNN regression algorithm.
2. Explain how K controls model flexibility.
3. Discuss the bias–variance tradeoff in KNN.
4. Explain the curse of dimensionality.
5. Explain why KNN performs poorly in high dimensions.
6. Compare parametric vs nonparametric regression.
7. Explain how kernel smoothing works.
8. Explain the role of distance metrics in KNN.
9. Explain why larger datasets benefit KNN.
10. Discuss advantages and disadvantages of KNN.

Coding Problems

Datasets: - Bike Sharing - California Housing

1. Implement KNN regression for predicting bike rentals.
2. Evaluate prediction error for $K = 1, 3, 5, 10, 20$.
3. Plot prediction error vs K.
4. Compare KNN with linear regression.
5. Perform train/test split.
6. Standardize predictors and compare results.

7. Evaluate KNN using cross-validation.
8. Visualize predictions vs actual values.
9. Investigate effect of adding more predictors.
10. Summarize best value of K.

Week 4: Classification Methods

Theory Problems

1. Explain logistic regression.
2. Derive the logistic function

$$P(Y = 1|X) = \frac{1}{1 + \exp\left(-\beta_0 - \beta^T X\right)}$$

3. Explain decision boundaries.
4. Explain LDA assumptions.
5. Compare logistic regression and LDA.
6. Explain Bayes classifier.
7. Explain classification error.
8. Explain ROC curves.
9. Explain precision vs recall.
10. Explain confusion matrix.

Coding Problems

Datasets: - Heart Disease - Breast Cancer

1. Fit logistic regression.
2. Compute classification accuracy.
3. Compute confusion matrix.
4. Fit LDA.
5. Compare logistic vs LDA.
6. Fit decision tree.
7. Visualize tree.
8. Compute ROC curve.
9. Compare classifiers.
10. Evaluate using cross-validation.

Week 5: Ensemble Methods

Theory Problems

1. Explain bagging.
2. Explain random forests.
3. Explain boosting.
4. Explain variance reduction.
5. Explain feature importance.
6. Compare trees vs forests.
7. Explain bias in boosting.
8. Explain gradient boosting intuition.
9. Explain overfitting in ensembles.
10. Explain out-of-bag error.

Coding Problems

Datasets: - Titanic - Adult Income

1. Fit decision tree.
2. Fit random forest.
3. Compute accuracy.
4. Compute feature importance.
5. Fit boosting model.
6. Compare performance.
7. Tune number of trees.
8. Evaluate OOB error.
9. Plot feature importance.
10. Compare ensemble vs single model.

Week 6: Unsupervised Learning

Theory Problems

1. Explain clustering.
2. Explain K-means algorithm.
3. Explain PCA.
4. Explain eigenvectors in PCA.
5. Explain variance explained.
6. Explain dimensionality reduction.

7. Explain cluster evaluation.
8. Compare clustering vs classification.
9. Explain hierarchical clustering.
10. Explain PCA visualization.

Coding Problems

Datasets: - Mall Customers - Wholesale Customers

1. Apply K-means clustering.
2. Choose optimal K using elbow method.
3. Visualize clusters.
4. Apply PCA.
5. Plot first two principal components.
6. Compare clustering before/after PCA.
7. Apply hierarchical clustering.
8. Visualize dendrogram.
9. Interpret clusters.
10. Write a short summary.

Instructions

- Show all mathematical derivations clearly.
- For simulation questions, set a random seed and report your findings with brief interpretations.
- For coding problems, include both code and short written conclusions.
- Unless otherwise stated, use **R**.
- You may use built-in R functions, but if a problem asks for a manual implementation, do not rely entirely on a black-box function.

Problem 1: Bias-Variance Decomposition

Suppose the data are generated from the model

$$Y = f(X) + \varepsilon,$$

where

$$E(\varepsilon \mid X) = 0, \qquad \mathrm{Var}(\varepsilon \mid X) = \sigma^2.$$

Let $\hat{f}(x_0)$ be an estimator of $f(x_0)$ based on a training sample, and let

$$Y_0 = f(x_0) + \varepsilon_0$$

be a future response at the same predictor value x_0.

1. Show that the expected prediction error at x_0 is

$$E\big[(Y_0 - \hat{f}(x_0))^2\big].$$

2. Derive the decomposition

$$E\big[(Y_0 - \hat{f}(x_0))^2\big] = \mathrm{Var}(\hat{f}(x_0)) + \big(E[\hat{f}(x_0)] - f(x_0)\big)^2 + \sigma^2.$$

3. Clearly explain the meaning of the three terms in this decomposition.

Problem 2: OLS Unbiasedness and Variance

Consider the linear model

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon,$$

where

$$E(\varepsilon) = \mathbf{0}, \qquad \mathrm{Var}(\varepsilon) = \sigma^2\mathbf{I}.$$

Assume $\mathbf{X}$ has full column rank.

1. Show that the ordinary least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.$$

2. Prove that $\hat{\beta}$ is unbiased.
3. Show that

$$\mathrm{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

4. Briefly explain why highly correlated predictors can inflate the variance of $\hat{\beta}$.

Problem 3: Training Error versus Testing Error

Suppose two models are fitted on the same training set:

- Model A: simple linear regression
- Model B: polynomial regression of degree 10

Answer the following.

1. Which model is guaranteed to have smaller or equal **training RSS**? Explain.
2. Which model is more likely to have higher variance? Explain.
3. Under what circumstance might Model B have larger test error even if its training error is smaller?
4. Explain how this connects to overfitting.

Problem 4: Effect of Correlation on OLS Stability

Consider a regression model with two predictors x_1 and x_2.

1. Suppose x_1 and x_2 are highly correlated. Explain why the fitted coefficients may become unstable.
2. Explain why unstable coefficients do not necessarily imply poor training fit.
3. In such a setting, explain why prediction may still be reasonable even if individual coefficient estimates are highly variable.

Problem 5: Parametric versus Nonparametric Models

Answer the following.

1. Explain the difference between a parametric model and a nonparametric model.
2. Give one example of each.
3. Compare them in terms of:
   - flexibility,
   - interpretability,
   - sample size requirement,
   - risk of overfitting.

Problem 6: Simulation of OLS under a Correctly Specified Model

In this problem, you will examine the sampling variability of OLS.

Generate data from

$$y_i = 3 + 2x_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 1),$$

with $n = 100$ and $x_i \sim \text{Unif}(0, 1)$.

1. Write R code to generate one dataset and fit the model $y \sim x$.
2. Repeat the simulation 500 times.
3. Store the estimated slope and intercept from each repetition.
4. Plot histograms of the 500 estimated slopes and intercepts.
5. Compute the empirical mean and empirical standard deviation of the estimated slope.
6. Compare the empirical mean of the slope estimator with the true value 2.
7. Briefly comment on whether the simulation agrees with the theoretical unbiasedness of OLS.

Problem 7: Simulation under Model Misspecification

Now generate data from

$$y_i = 3 + 2x_i + x_i^2 + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 1),$$

with $n = 100$ and $x_i \sim \text{Unif}(0, 1)$.

However, fit the misspecified model

$$y \sim x.$$

1. Simulate one dataset and fit the linear model.
2. Repeat the experiment 500 times.

3. Plot the distribution of the estimated slope.
4. Compare the average fitted slope with the true linear coefficient 2.
5. Explain why the fitted linear model may still show systematic bias even when OLS is unbiased for the best linear approximation.

Problem 8: Real Data Analysis with the Auto Dataset

Use the `Auto` dataset from the `ISLR2` package.

1. Load the data and remove observations with missing values if necessary.
2. Fit the simple linear regression model

$$\texttt{mpg} \sim \texttt{horsepower}.$$

3. Report:
   - estimated coefficients,
   - R^2,
   - residual standard error.
4. Plot the fitted regression line on top of the scatterplot.
5. Based on the plot, comment on whether a linear relationship seems adequate.

Problem 9: Polynomial Regression and Model Complexity

Continue using the `Auto` dataset.

Fit polynomial regression models of degree $1, 2, 3, 4, 5$ for predicting `mpg` from `horsepower`.

1. For each degree, report the training MSE.
2. Randomly split the data into a training set (70%) and a testing set (30%).
3. For each degree, compute the testing MSE.
4. Make one plot of degree versus training MSE.
5. Make another plot of degree versus testing MSE.
6. Explain how the two plots illustrate the bias-variance trade-off.

Problem 10: Cross-Validation and Model Selection

Again using the `Auto` dataset, compare polynomial models of degree 1 through 10 using **10-fold cross-validation**.

1. Use cross-validation to estimate prediction error for each degree.
2. Identify the degree with the smallest CV error.
3. Compare your selected degree with the model selected by the test-set approach in Problem 9.
4. Explain why the two approaches may not choose exactly the same model.

Problem 11: Diagnostics and Interpretation

Using your preferred polynomial model from Problems 9–10:

1. Plot residuals versus fitted values.
2. Produce a normal Q-Q plot of the residuals.

3. Comment on:
    - possible nonlinearity,
    - nonconstant variance,
    - departures from normality.
4. State whether these diagnostics suggest that your chosen model is reasonable.

Problem 12: Short Reflection

Write a short paragraph addressing the following questions.

1. What is the main difference between **training performance** and **generalization performance**?
2. Why is a more flexible model not always better?
3. What did the simulations teach you about estimator variability?

Optional Challenge

Create your own simulation where the true regression function is nonlinear but smooth, for example

$$y = \sin(2\pi x) + \varepsilon.$$

Compare:

- linear regression,
- polynomial regression,
- a nonparametric smoother of your choice.

Summarize which method gives the best prediction performance and why.

General Instructions

- Show all derivations clearly.
- For simulation questions, set a random seed and summarize your findings in words.
- For coding problems, include both code and interpretation.
- When a problem asks for a **manual implementation**, do not use only a black-box function.
- Unless otherwise stated, use **R**.

Week 1: Foundations, Bias-Variance Trade-off, and Linear Regression

Problem 1: Bias-Variance Decomposition

Suppose

$$Y = f(X) + \varepsilon,$$

where

$$E(\varepsilon \mid X) = 0, \qquad \mathrm{Var}(\varepsilon \mid X) = \sigma^2.$$

Let $\hat{f}(x_0)$ be an estimator of $f(x_0)$ based on a training sample, and let

$$Y_0 = f(x_0) + \varepsilon_0$$

be a future response at the predictor value x_0.

1. Show that the expected prediction error at x_0 is

$$E\big[(Y_0 - \hat{f}(x_0))^2\big].$$

2. Derive the decomposition

$$E\big[(Y_0 - \hat{f}(x_0))^2\big] = \mathrm{Var}(\hat{f}(x_0)) + \big(E[\hat{f}(x_0)] - f(x_0)\big)^2 + \sigma^2.$$

3. Explain the meaning of each term.

Problem 2: OLS Unbiasedness

Consider the linear model

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon,$$

where

$$E(\varepsilon) = \mathbf{0}, \qquad \mathrm{Var}(\varepsilon) = \sigma^2\mathbf{I}.$$

Assume $\mathbf{X}$ has full column rank.

1. Show that the OLS estimator is

$$\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.$$

2. Prove that $\hat{\beta}$ is unbiased.
3. Show that

$$\mathrm{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

Problem 3: Correlated Predictors and Variance Inflation

Suppose two predictors x_1 and x_2 are highly correlated.

1. Explain why the matrix $\mathbf{X}^\top\mathbf{X}$ becomes nearly singular.
2. Explain why this increases the variance of $\hat{\beta}$.
3. Explain why coefficient instability does not necessarily imply poor training fit.

Problem 4: Model Complexity

Suppose two models are fitted to the same dataset:

- Model A: simple linear regression
- Model B: polynomial regression of degree 10

Answer the following.

1. Which model must have smaller or equal training RSS?
2. Which model is likely to have higher variance?
3. Under what conditions could Model B have larger test error?

Problem 5: Parametric versus Nonparametric Models

1. Explain the difference between a parametric model and a nonparametric model.
2. Give one example of each.
3. Compare them in terms of flexibility, interpretability, and sample size requirement.

Problem 6: Simulation of OLS under a Correctly Specified Model

Generate data from

$$y_i = 3 + 2x_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 1),$$

with $n = 100$ and $x_i \sim \text{Unif}(0, 1)$.

1. Generate one dataset and fit the model $y \sim x$.
2. Repeat the simulation 500 times.
3. Store the estimated slope and intercept from each repetition.
4. Plot histograms of the estimated slopes and intercepts.
5. Compare the empirical mean of the estimated slope with the true value 2.

```r
set.seed(432)
# Your code here
```

Problem 7: Simulation under Misspecification

Now generate data from

$$y_i = 3 + 2x_i + x_i^2 + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 1),$$

with $n = 100$ and $x_i \sim \text{Unif}(0, 1)$, but still fit the linear model

$$y \sim x.$$

1. Repeat the experiment 500 times.
2. Plot the distribution of the estimated slope.
3. Compare the average fitted slope with the true linear coefficient 2.
4. Explain the result.

```r
set.seed(432)
# Your code here
```

Problem 8: Auto Dataset — Simple Linear Regression

Use the `Auto` dataset from `ISLR2`.

1. Fit the model

$$\texttt{mpg} \sim \texttt{horsepower}.$$

2. Report the estimated coefficients, R^2, and residual standard error.
3. Plot the data and add the fitted regression line.
4. Comment on whether the linear fit seems adequate.

Problem 9: Polynomial Regression and Test Error

Using the `Auto` dataset:

1. Randomly split the data into training (70%) and testing (30%).
2. Fit polynomial models of degree $1, \dots, 5$ predicting `mpg` from `horsepower`.
3. Compute the training MSE and testing MSE for each degree.
4. Plot degree versus training MSE and degree versus testing MSE.
5. Explain the pattern.

```r
set.seed(432)
# Your code here
```

Problem 10: Cross-Validation and Model Selection

Again using the `Auto` dataset:

1. Use 10-fold cross-validation to compare polynomial models of degree $1, \dots, 10$.
2. Identify the degree with the smallest CV error.
3. Compare your answer with Problem 9.
4. Explain why the chosen degree may differ between the test-set approach and cross-validation.

```r
set.seed(432)
# Your code here
```

Week 2: Ridge Regression, Lasso, and Regularization

Problem 1: Ridge Regression Formula

Ridge regression solves

$$\hat{\beta}^{ridge} = \arg\min_{\beta} \left\{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + n\lambda\|\beta\|_2^2 \right\}.$$

1. Show that the ridge solution is

$$\hat{\beta}^{ridge} = (\mathbf{X}^\top\mathbf{X} + n\lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}.$$

2. Explain why this always exists when $\lambda > 0$.

Problem 2: Ridge Variance Reduction

Assume the linear model from Week 1.

1. Explain why adding $n\lambda\mathbf{I}$ increases the eigenvalues of $\mathbf{X}^\top\mathbf{X}$.
2. Explain why this reduces estimator variance.
3. Explain why this introduces bias.

Problem 3: Bias-Variance Trade-off in Ridge

1. Explain what happens when $\lambda = 0$.
2. Explain what happens as $\lambda \to \infty$.
3. Explain why an intermediate value of λ may improve prediction.

Problem 4: Geometric Interpretation

1. Write the constrained form of ridge regression.
2. Explain the geometry of the L_2 ball.
3. Explain how this geometry leads to shrinkage but usually not exact zeros.

Problem 5: Lasso Geometry

The lasso solves

$$\hat{\beta}^{lasso} = \arg\min_{\beta} \left\{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + n\lambda\|\beta\|_1 \right\}.$$

1. Write the constrained form of lasso.
2. Explain the geometry of the L_1 ball.
3. Explain why lasso can set coefficients exactly to zero.

Problem 6: Standardization

1. Explain why predictors should be standardized before applying ridge or lasso.
2. Explain what could go wrong if they are not standardized.

Problem 7: Manual Ridge Prediction

Use the first 24 observations of `mtcars` as training data and the remaining observations as testing data.

1. Fit a ridge regression model with $\lambda = 3$.
2. Compute predictions on the test data manually using

$$\hat{y} = \mathbf{x}^\top \hat{\beta}.$$

3. Report the test MSE.

```r
library(MASS)
traindata <- mtcars[1:24, ]
testdata <- mtcars[-(1:24), ]
# Your code here
```

Problem 8: Coefficient Paths in Ridge

Using `mtcars` or `Hitters`:

1. Fit ridge regression for a grid of λ values.
2. Plot the coefficient paths.
3. Explain which coefficients shrink fastest and why that may happen.

```r
library(MASS)
# Your code here
```

Problem 9: Cross-Validation for Ridge

Using `Hitters` from `ISLR2`:

1. Remove missing values.
2. Use 10-fold cross-validation to select λ for ridge regression.
3. Report λ_{min}.
4. Report the corresponding CV error.

Problem 10: Ridge versus Lasso on Real Data

Using the same dataset:

1. Fit ridge regression and lasso with cross-validation.
2. Compare their test errors using a train/test split.
3. For lasso, report the selected variables.
4. Write a brief discussion comparing prediction and interpretability.

```r
set.seed(432)
# Your code here
```

Week 3: KNN, Flexible Regression, and Nonparametric Methods

Problem 1: KNN Regression Estimator

The KNN regression estimator at x_0 is

$$\hat{f}(x_0) = \frac{1}{K} \sum_{i \in N_K(x_0)} y_i,$$

where $N_K(x_0)$ is the set of indices of the K nearest training points.

1. Explain how this estimator works.
2. Explain why $K = 1$ gives a very flexible fit.
3. Explain why large K gives a smoother fit.

Problem 2: Bias-Variance in KNN

1. Explain how increasing K changes bias.
2. Explain how increasing K changes variance.
3. Explain why there is usually an optimal value of K.

Problem 3: Curse of Dimensionality

1. Explain why nearest neighbors become less meaningful in high dimensions.
2. Explain why nonparametric methods often require more data than parametric methods.

Problem 4: Distance Metrics

1. Explain why standardization matters in KNN.
2. Explain how one predictor with a much larger scale can distort the neighborhood structure.

Problem 5: Comparison with Linear Regression

1. Give an example where linear regression is preferable to KNN.
2. Give an example where KNN is preferable to linear regression.

Problem 6: Simulated Nonlinear Regression

Generate data from

$$y = \sin(2\pi x) + \varepsilon, \qquad \varepsilon \sim N(0, 0.2^2).$$

1. Fit linear regression.
2. Fit polynomial regression of degree 5.
3. Fit KNN regression for several values of K.
4. Compare test MSE.

```
set.seed(432)
# Your code here
```

Problem 7: KNN on the Bike Sharing Data

Use the Bike Sharing dataset.

1. Select a response variable related to rental counts.
2. Split the data into training and testing sets.
3. Fit KNN regression for $K = 1, 3, 5, 10, 20, 50$.
4. Plot test MSE versus K.
5. Identify the best K.

```
# Your code here
```

Problem 8: Effect of Standardization in KNN

Using the same data:

1. Fit KNN without standardization.
2. Fit KNN with standardization.
3. Compare the results and explain the difference.

```
# Your code here
```

Problem 9: KNN versus Linear Regression

Using the same train/test split:

1. Fit a linear regression model.
2. Compare its test MSE with the best KNN model.
3. Explain which model appears more suitable and why.

```
# Your code here
```

Problem 10: Smoothing Interpretation

Use a one-dimensional predictor/response pair from any open dataset.

1. Fit KNN or another smoother.
2. Plot the fitted curve.
3. Explain how the method captures nonlinearity.

```
# Your code here
```

Week 4: Classification, Logistic Regression, LDA, and Trees

Problem 1: Logistic Regression Model

For binary classification, logistic regression assumes

$$P(Y = 1 \mid X) = \frac{1}{1 + \exp(-\beta_0 - \beta^\top X)}.$$

1. Explain why this probability is always between 0 and 1.
2. Show that the log-odds are linear in X:

$$\log \frac{P(Y = 1 \mid X)}{1 - P(Y = 1 \mid X)} = \beta_0 + \beta^\top X.$$

Problem 2: Classification Rule

Suppose a fitted logistic model gives

$$\hat{P}(Y = 1 \mid X = x) = 0.73.$$

1. Classify the observation using the threshold 0.5.

2. Explain how the classification changes if the threshold is 0.8.

Problem 3: LDA Assumptions

LDA assumes that within each class the predictors are multivariate normal with class-specific means and a common covariance matrix.

1. Explain why this leads to a linear decision boundary.
2. Explain when LDA may outperform logistic regression.

Problem 4: Bayes Classifier

1. Explain the idea of the Bayes classifier.
2. Explain why it is theoretically optimal.
3. Explain why it is not directly available in practice.

Problem 5: Trees

1. Explain how a decision tree recursively partitions the predictor space.
2. Explain why the prediction regions are piecewise constant.
3. Explain one advantage and one disadvantage of trees.

Problem 6: Heart Disease Data — Logistic Regression

Use the UCI Heart Disease dataset or another open binary classification dataset.

1. Split the data into training and testing sets.
2. Fit a logistic regression model.
3. Report test accuracy, sensitivity, and specificity.
4. Report the confusion matrix.

```
# Your code here
```

Problem 7: Logistic Regression Interpretation

Using your fitted logistic model:

1. Select one numerical predictor.
2. Interpret its coefficient in terms of odds ratio.
3. Comment on whether the sign matches your intuition.

```
# Your code here
```

Problem 8: LDA Comparison

Using the same train/test split:

1. Fit an LDA classifier.
2. Compare its accuracy with logistic regression.

3. Comment on which method performs better and why that might happen.

Problem 9: Decision Tree

Using the same dataset:

1. Fit a classification tree.
2. Plot the tree.
3. Report test accuracy.
4. Compare with logistic regression and LDA.

```
# Your code here
```

Problem 10: ROC Curve and Thresholding

Using the logistic regression model:

1. Compute predicted probabilities on the test set.
2. Draw the ROC curve.
3. Compute AUC.
4. Explain why accuracy alone may not fully describe classifier performance.

```
# Your code here
```

Week 5: Bagging, Random Forests, and Boosting

Problem 1: Bagging and Variance Reduction

1. Explain why a single tree often has high variance.
2. Explain how bagging reduces variance.
3. Explain why bagging does not usually reduce bias very much.

Problem 2: Random Forests

1. Explain how random forests modify bagging.
2. Explain why selecting only a random subset of predictors at each split helps.
3. Explain what out-of-bag error is.

Problem 3: Boosting

1. Explain the basic idea of boosting.
2. Explain why later trees focus on difficult observations.
3. Explain how boosting can reduce bias.

Problem 4: Feature Importance

1. Explain how variable importance is measured in random forests.
2. Explain why importance should be interpreted carefully.

Problem 5: Trees versus Ensembles

Compare a single tree and a random forest in terms of:

1. interpretability,
2. variance,
3. prediction accuracy.

Problem 6: Titanic Data — Single Tree

Use the Titanic dataset.

1. Fit a classification tree predicting survival.
2. Report training accuracy and test accuracy.
3. Plot the tree.
4. Briefly interpret the first two splits.

```
# Your code here
```

Problem 7: Titanic Data — Random Forest

Using the same train/test split:

1. Fit a random forest.
2. Report test accuracy.
3. Report out-of-bag error.
4. Plot variable importance.

```
# Your code here
```

Problem 8: Random Forest Tuning

1. Fit random forests with different values of `mtry`.
2. Compare out-of-bag error across settings.
3. Identify the best choice of `mtry`.

```
# Your code here
```

Problem 9: Boosting

Using the same dataset or another open classification dataset:

1. Fit a boosting model.
2. Tune the number of trees.
3. Report test accuracy.
4. Compare with the tree and the random forest.

```
# Your code here
```

Problem 10: Ensemble Comparison

Summarize the performance of:

- a single tree,
- bagging or random forest,
- boosting.

Discuss which method performs best and why.

```
# Your code here
```

Week 6: Unsupervised Learning, PCA, and Clustering

Problem 1: K-means Clustering

1. Describe the K-means algorithm.
2. Explain the objective function minimized by K-means.
3. Explain why different initializations can lead to different results.

Problem 2: PCA

1. Explain the goal of PCA.
2. Explain what the first principal component represents.
3. Explain why PCA is useful before clustering or visualization.

Problem 3: Variance Explained

1. Explain the proportion of variance explained by each principal component.
2. Explain how to decide how many principal components to keep.

Problem 4: Clustering versus Classification

1. Explain the difference between unsupervised clustering and supervised classification.
2. Explain why clustering is generally harder to evaluate.

Problem 5: Standardization in PCA and Clustering

1. Explain why standardization is often important before PCA.
2. Explain why it is often important before K-means.

Problem 6: Mall Customers — K-means

Use the Mall Customers dataset or another open clustering dataset.

1. Standardize the variables.
2. Fit K-means with $K = 2, \dots, 8$.
3. Plot the within-cluster sum of squares versus K.
4. Choose a reasonable value of K and justify your choice.

```
# Your code here
```

Problem 7: PCA Visualization

Using the same dataset:

1. Perform PCA.
2. Plot the first two principal components.
3. Color the points by the K-means cluster labels.
4. Comment on whether the clusters appear well separated.

```
# Your code here
```

Problem 8: Clustering Before and After PCA

1. Cluster the original standardized data.
2. Cluster the data using only the first two principal components.
3. Compare the cluster assignments.
4. Discuss whether PCA helped.

```
# Your code here
```

Problem 9: Hierarchical Clustering

Using the same data:

1. Perform hierarchical clustering.
2. Plot the dendrogram.
3. Cut the tree to obtain a chosen number of clusters.
4. Compare with the K-means result.

```
# Your code here
```

Problem 10: Final Reflection

Write a short reflection addressing the following.

1. Which supervised method did you find easiest to interpret?
2. Which method seemed to give the best predictive performance?
3. What are the main challenges of unsupervised learning?

Optional Challenge Problems

Challenge 1: Manual Cross-Validation

Write your own function for K-fold cross-validation for one of the following:

- ridge regression,
- lasso,
- KNN.

Do not use a built-in CV wrapper.

```
# Your code here
```

Challenge 2: Simulation Study of the Curse of Dimensionality

Simulate a nonlinear regression setting and compare KNN performance as the number of predictors increases from $p = 1$ to $p = 20$.

```
# Your code here
```

Challenge 3: Final Mini-Project

Choose one open dataset and compare at least three methods from the course. Include:

- train/test split,
- model tuning,
- performance comparison,
- interpretation.

```
# Your code here
```